

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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**in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING in COMPUTER SCIENCE AND
ENGINEERING**



**B.M.S. COLLEGE OF ENGINEERING (Autonomous Institution under
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**B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum) Department
of Computer Science and Engineering**



CERTIFICATE

This is to certify that the Lab work entitled “**ANALYSIS AND DESIGN OF ALGORITHMS**” carried out by **Mandeep Singh(1BF24CS166)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab Program 1:

Write program to obtain the Topological ordering of vertices in a given digraph.

Code:

```
#include <stdio.h>

#define MAX 10

int main() {
    int n;
    int graph[MAX][MAX];
    int indegree[MAX] = {0};
    int visited[MAX] = {0};

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter adjacency matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &graph[i][j]);

            if (graph[i][j] == 1)
                indegree[j]++;
        }
    }

    printf("\nTopological Ordering:\n");

    for (int count = 0; count < n; count++) {

        for (int i = 0; i < n; i++) {

            if (indegree[i] == 0 && !visited[i]) {

                printf("%d ", i);
                visited[i] = 1;

                // Remove outgoing edges
                for (int j = 0; j < n; j++) {
                    if (graph[i][j] == 1)
                        indegree[j]--;
                }

                break;
            }
        }
    }
}
```

```
    }  
  }  
}  
  
printf("\n");  
  
return 0;  
}
```

Output:

OutputClear

```
Enter number of vertices: 3  
Enter adjacency matrix:  
0 1 0  
0 0 0  
1 1 0  
  
Topological Ordering:  
2 0 1  
  
=== Code Execution Successful ===
```

LeetCode Problem 207:

There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [a_i, b_i] indicates that you must take course b_i first if you want to take course a_i.

- For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.

Return true if you can finish all courses. Otherwise, return false.

Code:

```
#include <stdlib.h>

#include <stdbool.h>

bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize) {
    int *indegree = (int *)calloc(numCourses, sizeof(int));
    int **adj = (int **)malloc(numCourses * sizeof(int *));
    int *adjSize = (int *)calloc(numCourses, sizeof(int));
    for (int i = 0; i < numCourses; i++) {
        adj[i] = (int *)malloc(numCourses * sizeof(int));
    }
    for (int i = 0; i < prerequisitesSize; i++) {
        int a = prerequisites[i][0];
        int b = prerequisites[i][1];
        adj[b][adjSize[b]++] = a;
        indegree[a]++;
    }
    int *queue = (int *)malloc(numCourses * sizeof(int));
    int front = 0, rear = 0;
    for (int i = 0; i < numCourses; i++) {
        if (indegree[i] == 0) {
            queue[rear++] = i;
        }
    }
```

```

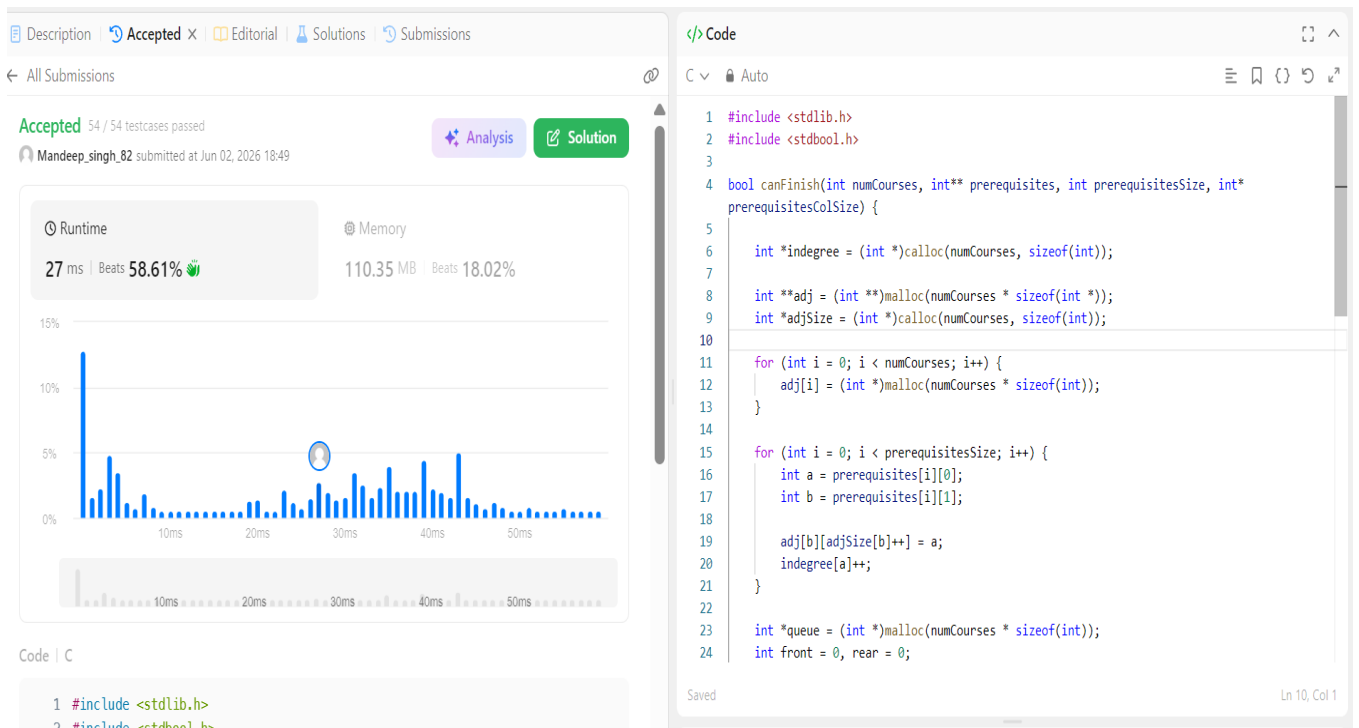
    }

    int count = 0;
    while (front < rear) {
        int course = queue[front++];
        count++;
        for (int i = 0; i < adjSize[course]; i++) {
            int next = adj[course][i];
            indegree[next]--;
            if (indegree[next] == 0) {
                queue[rear++] = next;
            }
        }
    }

    for (int i = 0; i < numCourses; i++) {
        free(adj[i]);
    }
    free(adj);
    free(adjSize);
    free(indegree);
    free(queue);
    return count == numCourses;
}

```

ScreenShot from LeetCode:



Lab Program 2:

Implement Johnson Trotter algorithm to generate permutations.

Code:

```
#include <stdio.h>

#define MAX 10

#define L -1

#define R 1

void print(int a[], int n) {
    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
}

int main() {
    int n, a[MAX], dir[MAX];
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        a[i] = i + 1;
        dir[i] = L;
    }
    print(a, n);
    while (1) {
        int mobile = 0, pos = -1;
        for (int i = 0; i < n; i++) {
            if (dir[i] == L && i > 0 &&
                a[i] > a[i-1] && a[i] > mobile) {
                mobile = a[i];
                pos = i;
            }
        }
    }
}
```

```

    if (dir[i] == R && i < n-1 &&
        a[i] > a[i+1] && a[i] > mobile) {
        mobile = a[i];
        pos = i;
    }
}

if (mobile == 0)
    break;

int next = pos + dir[pos];
int t = a[pos];
a[pos] = a[next];
a[next] = t;
t = dir[pos];
dir[pos] = dir[next];
dir[next] = t;

for (int i = 0; i < n; i++)
    if (a[i] > mobile)
        dir[i] *= -1;

print(a, n);
}

return 0;
}

```

Output:

```
Output Clear  
Enter n: 3  
1 2 3  
1 3 2  
3 1 2  
3 2 1  
2 3 1  
2 1 3  
  
=== Code Execution Successful ===
```

Lab Program:3

Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

Code:

```
#include <stdio.h>

#include <time.h>

#define MAX 100000

void merge(int a[], int low, int mid, int high)
{
    int i = low, j = mid + 1, k = low;
    int temp[MAX];
    while (i <= mid && j <= high)
    {
        if (a[i] <= a[j])
            temp[k++] = a[i++];
        else
            temp[k++] = a[j++];
    }
    while (i <= mid)
        temp[k++] = a[i++];
    while (j <= high)
        temp[k++] = a[j++];
    for (i = low; i <= high; i++)
        a[i] = temp[i];
}

void mergeSort(int a[], int low, int high)
{
    if (low < high)
    {

```

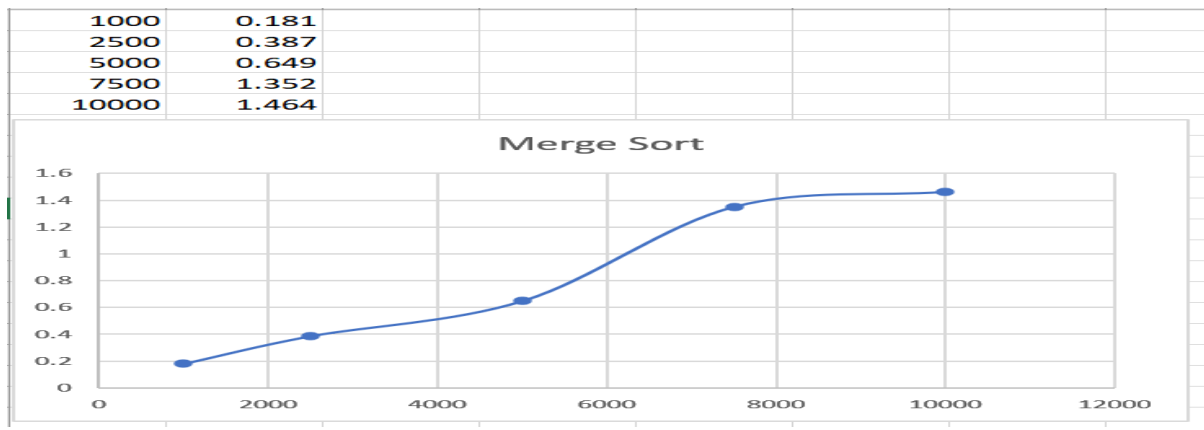
```

        int mid = (low + high) / 2;
        mergeSort(a, low, mid);
        mergeSort(a, mid + 1, high);
        merge(a, low, mid, high);
    }
}

int main()
{
    int a[MAX], n, i;
    clock_t start, end;
    double time_taken;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter elements:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    start = clock();
    mergeSort(a, 0, n - 1);
    end = clock();
    time_taken = ((double)(end - start)*1000.0) / CLOCKS_PER_SEC;
    printf("\nSorted array:\n");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n\nTime taken = %f seconds\n", time_taken);
    return 0;
}

```

Output:



Enter number of elements: 5

Enter elements:

41 5 78 49 30

Sorted array:

5 30 41 49 78

Time taken = 0.014000 seconds

LeetCode Problem 21:

You are given the heads of two sorted linked lists list1 and list2.

Merge the two lists into one **sorted** list. The list should be made by splicing together the nodes of the first two lists.

Return *the head of the merged linked list*.

Code:

```
/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     struct ListNode *next;
 * };
 */

struct ListNode* mergeTwoLists(struct ListNode* list1, struct ListNode* list2) {
    struct ListNode *temp=list1;
    struct ListNode *i,*j;
    int temp2;
    if(list1==NULL){
        list1=list2;
        return list1;
    }
    while(temp->next != NULL){
        temp=temp->next;
    }
    temp->next=list2;
    for(i=list1;i!=NULL;i=i->next){
        for(j=i->next;j!=NULL;j=j->next){
```

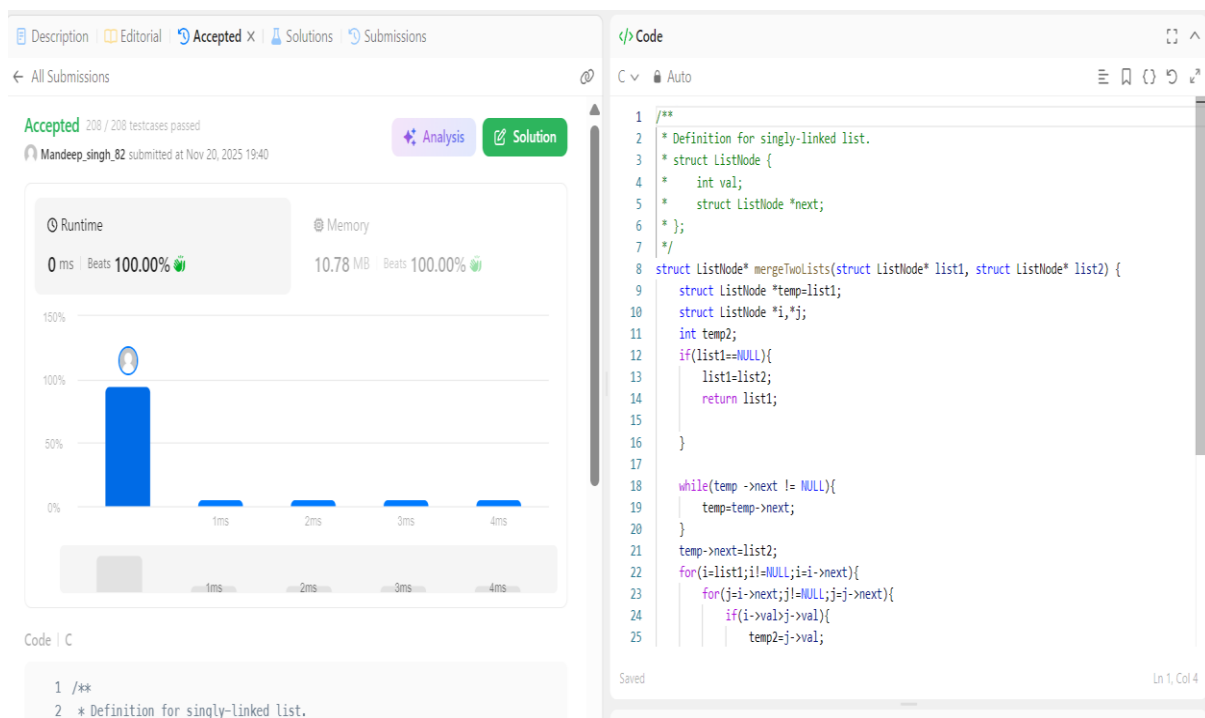
```

        if(i->val>j->val){
            temp2=j->val;
            j->val=i->val;
            i->val=temp2;
        }
    }
}

return list1;
}

```

ScreenShot from Leetcode:



Lab Program:4

Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

void swap(int* a, int* b) {
    int t = *a;
    *a = *b;
    *b = t;
}

int partition(int arr[], int low, int high) {
    int pivot = arr[low];
    int i = low;
    int j = high;

    while (i < j) {

        while (arr[i] <= pivot && i <= high - 1) {
            i++;
        }

        while (arr[j] > pivot && j >= low + 1) {
            j--;
        }

        if (i < j) {
            swap(&arr[i], &arr[j]);
        }
    }

    swap(&arr[low], &arr[j]);
    return j;
}

void quickSort(int arr[], int low, int high) {
```

```

    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}

int main() {
    int n;
    printf("Enter number of elements: ");
    if (scanf("%d", &n) != 1) return 1;

    int *arr = (int*)malloc(n * sizeof(int));

    srand(time(0));
    for (int i = 0; i < n; i++) {
        arr[i] = rand() % 10000;
    }

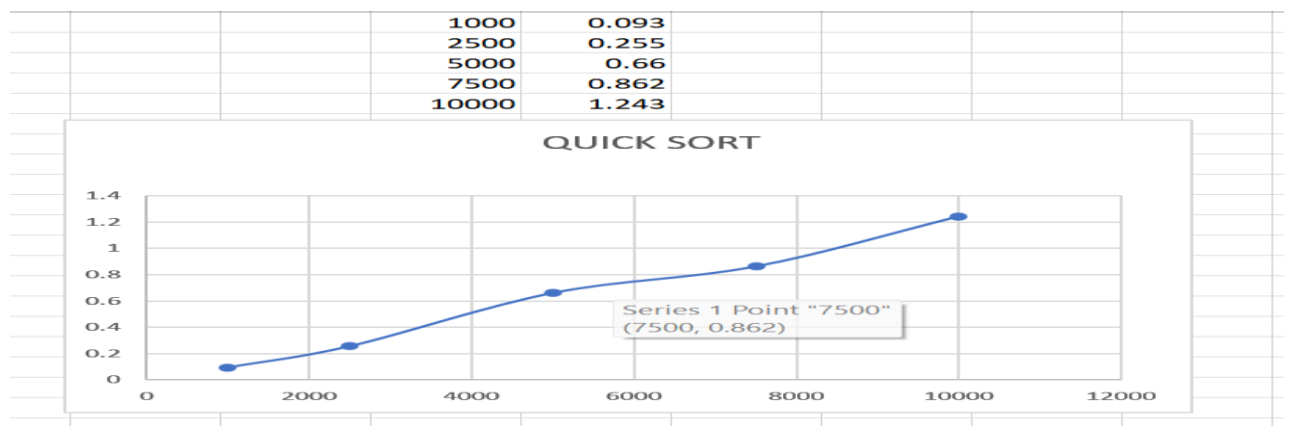
    clock_t start = clock();
    quickSort(arr, 0, n - 1);
    clock_t end = clock();

    double cpu_time_used = ((double)(end - start)) / CLOCKS_PER_SEC;

    printf("\nSorting complete.\n");
    printf("Time taken: %f seconds\n", cpu_time_used);
}

```

Output:



Output

Clear

Enter number of elements: 5

Enter 5 elements

51 46 79 64 31

Sorting complete.

31 46 51 64 79

Time taken: 0.000002 seconds

=== Code Execution Successful ===

LeetCode Problem:997

In a town, there are n people labeled from 1 to n . There is a rumor that one of these people is secretly the town judge.

If the town judge exists, then:

1. The town judge trusts nobody.
2. Everybody (except for the town judge) trusts the town judge.
3. There is exactly one person that satisfies properties **1** and **2**.

You are given an array `trust` where `trust[i] = [ai, bi]` representing that the person labeled a_i trusts the person labeled b_i . If a trust relationship does not exist in trust array, then such a trust relationship does not exist.

Return *the label of the town judge if the town judge exists and can be identified, or return -1 otherwise.*

Code:

```
int findJudge(int n, int** trust, int trustSize, int* trustColSize) {  
    int count[n + 1];  
  
    for (int i = 0; i <= n; i++)  
        count[i] = 0;  
  
    for (int i = 0; i < trustSize; i++) {  
        int a = trust[i][0];  
        int b = trust[i][1];  
  
        count[a]--;  
        count[b]++;  
    }  
  
    for (int i = 1; i <= n; i++) {  
        if (count[i] == n - 1)  
            return i;  
    }  
}
```

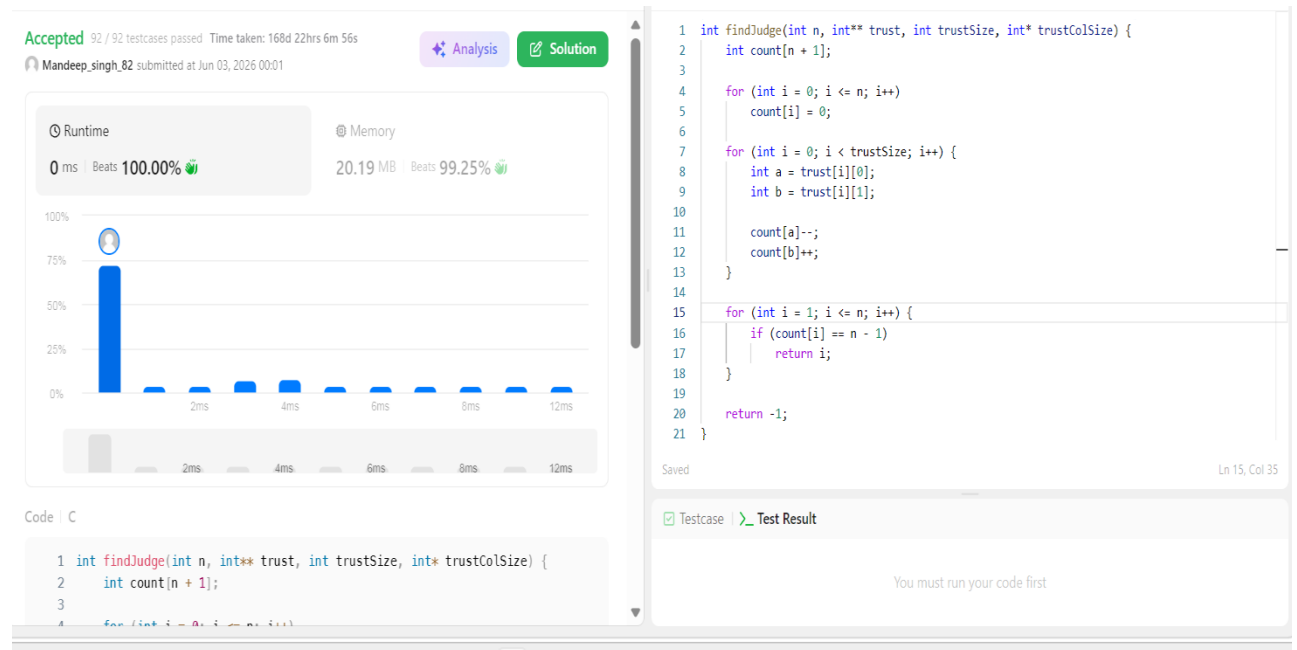
```

return -1;

}

```

ScreenShot from Leetcode:



Lab Program:5

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

Code:

```
#include <stdio.h>
#include <time.h>

void heapify(int a[], int n, int i) {
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;

    if (left < n && a[left] > a[largest])
        largest = left;

    if (right < n && a[right] > a[largest])
        largest = right;

    if (largest != i) {
        int temp = a[i];
        a[i] = a[largest];
        a[largest] = temp;

        heapify(a, n, largest);
    }
}

void heapSort(int a[], int n) {
    for (int i = n/2 - 1; i >= 0; i--)
        heapify(a, n, i);

    for (int i = n-1; i > 0; i--) {
        int temp = a[0];
        a[0] = a[i];
        a[i] = temp;

        heapify(a, i, 0);
    }
}
```

```

int main() {
    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int a[n];

    printf("Enter elements:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    clock_t start, end;

    start = clock();

    heapSort(a, n);

    end = clock();

    printf("\nSorted Array:\n");
    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);

    double time_taken =
        ((double)(end - start)) / CLOCKS_PER_SEC;

    printf("\n\nTime Taken = %f seconds\n", time_taken);

    return 0;
}

```

Output:

Output

Clear

Enter number of elements: 10

Enter elements:

6 15 76 8 0 57 12 90 24 1

Sorted Array:

0 1 6 8 12 15 24 57 76 90

Time Taken = 0.000002 seconds

=== Code Execution Successful ===

Lab Program:6

Implement 0/1 Knapsack problem using dynamic programming

Code:

```
#include <stdio.h>
int max(int a, int b) {
    return (a > b) ? a : b;
}
int main() {
    int n, W;
    printf("Enter number of items: ");
    scanf("%d", &n);

    int wt[n], val[n];
    printf("Enter weights:\n");

    for (int i = 0; i < n; i++)
        scanf("%d", &wt[i]);
    printf("Enter values:\n");

    for (int i = 0; i < n; i++)
        scanf("%d", &val[i]);

    printf("Enter capacity: ");
    scanf("%d", &W);

    int dp[n+1][W+1];

    for (int i = 0; i <= n; i++) {
        for (int w = 0; w <= W; w++) {

            if (i == 0 || w == 0)
                dp[i][w] = 0;

            else if (wt[i-1] <= w)
                dp[i][w] = max(
                    val[i-1] + dp[i-1][w - wt[i-1]],
                    dp[i-1][w]
                );
            else
                dp[i][w] = dp[i-1][w];
        }
    }
    printf("\nMaximum Value = %d\n", dp[n][W]);
    return 0;
}
```

Output:

```
Output Clear  
Enter number of items: 4  
Enter weights:  
50  
20  
30  
10  
Enter values:  
120  
200  
100  
110  
Enter capacity: 90  
  
Maximum Value = 430  
  
=== Code Execution Successful ===
```

LeetCode Problem:416

Given an integer array *nums*, return true *if you can partition the array into two subsets such that the sum of the elements in both subsets is equal* or false otherwise.

Code:

```
#include <stdbool.h>
#include <stdlib.h>

bool canPartition(int* nums, int numsSize) {
    int sum = 0;

    for (int i = 0; i < numsSize; i++) {
        sum += nums[i];
    }

    if (sum % 2 != 0)
        return false;

    int target = sum / 2;

    bool *dp = (bool *)calloc(target + 1, sizeof(bool));
    dp[0] = true;

    for (int i = 0; i < numsSize; i++) {
        for (int j = target; j >= nums[i]; j--) {
            dp[j] = dp[j] || dp[j - nums[i]];
        }
    }

    bool result = dp[target];
    free(dp);

    return result;
}
```

ScreenShot from LeetCode:

DescriptionAccepted XEditorialSolutionsSubmissions

← All Submissions

Accepted 147 / 147 testcases passed

Mandeep_singh_82 submitted at Jun 02, 2026 19:37

AnalysisSolution

Runtime47 msBeats 50.78%

Memory9.08 MBBeats 39.91%



Code | C

```
1 #include <stdbool.h>
2 #include <stdlib.h>
3
4 bool canPartition(int* nums, int numsSize) {
5     int sum = 0;
6
7     for (int i = 0; i < numsSize; i++) {
8         sum += nums[i];
9     }
10
11     if (sum % 2 != 0)
12         return false;
13
14     int target = sum / 2;
15
16     bool *dp = (bool *)calloc(target + 1, sizeof(bool));
17     dp[0] = true;
18
19     for (int i = 0; i < numsSize; i++) {
20         for (int j = target; j >= nums[i]; j--) {
21             dp[j] = dp[j] || dp[j - nums[i]];
22         }
23     }
24
25     bool result = dp[target];
26     free(dp);
27
28     return result;
29 }
```

</> Code

C v Auto

```
1 #include <stdbool.h>
2 #include <stdlib.h>
3
4 bool canPartition(int* nums, int numsSize) {
5     int sum = 0;
6
7     for (int i = 0; i < numsSize; i++) {
8         sum += nums[i];
9     }
10
11     if (sum % 2 != 0)
12         return false;
13
14     int target = sum / 2;
15
16     bool *dp = (bool *)calloc(target + 1, sizeof(bool));
17     dp[0] = true;
18
19     for (int i = 0; i < numsSize; i++) {
20         for (int j = target; j >= nums[i]; j--) {
21             dp[j] = dp[j] || dp[j - nums[i]];
22         }
23     }
24
25     bool result = dp[target];
26     free(dp);
27
28     return result;
29 }
```

SavedLn 27, Col 1

TestcaseTest Result

Lab Program:7

Implement All Pair Shortest paths problem using Floyd's algorithm

Code:

```
#include <stdio.h>
#define MAX 100
#define INF 99999

void floyd(int graph[MAX][MAX], int n) {
    int dist[MAX][MAX];

    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            dist[i][j] = graph[i][j];

    for (int k = 0; k < n; k++) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (dist[i][k] + dist[k][j] < dist[i][j]) {
                    dist[i][j] = dist[i][k] + dist[k][j];
                }
            }
        }
    }

    printf("\nAll-Pairs Shortest Path Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (dist[i][j] == INF)
                printf("INF ");
            else
                printf("%3d ", dist[i][j]);
        }
        printf("\n");
    }
}

int main() {
    int n, graph[MAX][MAX];

    printf("Enter number of vertices: ");
    scanf("%d", &n);
```

```
printf("Enter adjacency matrix (use 99999 for INF):\n");
for (int i = 0; i < n; i++)
    for (int j = 0; j < n; j++)
        scanf("%d", &graph[i][j]);

floyd(graph, n);

return 0;
}
```

Output:

Output

Clear

```
Enter number of vertices: 5
Enter adjacency matrix (use 99999 for INF):
0 5 6 9999 2
5 0 9999 4 6
6 9999 0 3 5
9999 4 3 0 4
2 6 5 4 0

All-Pairs Shortest Path Matrix:
0  5  6  6  2
5  0  7  4  6
6  7  0  3  5
6  4  3  0  4
2  6  5  4  0

=== Code Execution Successful ===
```

LeetCode Problem:783

Given the root of a Binary Search Tree (BST), return *the minimum difference between the values of any two different nodes in the tree.*

Code:

```
void inorder(struct TreeNode* root, int* prev, int* ans) {
    if (root == NULL)
        return;
    inorder(root->left, prev, ans);
    if (*prev != -1) {
        int diff = root->val - *prev;
        if (diff < *ans)
            *ans = diff;
    }
    *prev = root->val;
    inorder(root->right, prev, ans);
}

int minDiffInBST(struct TreeNode* root) {
    int prev = -1;
    int ans = 1000000;
    inorder(root, &prev, &ans);
    return ans;
}
```

ScreenShot from Leetcode:

Description

Accepted x

Editorial

Solutions

Submissions

All Submissions

Accepted 49 / 49 testcases passed

Mandeep_singh,82 submitted at Jun 03, 2026 00:09

Analysis

Solution

Runtime

0 ms | Beats 100.00%

Memory

9.59 MB | Beats 42.74%

The chart displays performance metrics for the submission. The runtime bar is at 0 ms, indicating it is the fastest. The memory bar is at 9.59 MB. The chart also shows a distribution of other submissions' performance, with a blue bar representing the user's performance and a grey bar representing the average.

Code | C

```
1 void inorder(struct TreeNode* root, int* prev, int* ans) {
2     if (root == NULL)
3         return;
4 
```

</> Code

C v Auto

```
1 void inorder(struct TreeNode* root, int* prev, int* ans) {
2     if (root == NULL)
3         return;
4     inorder(root->left, prev, ans);
5
6     if (*prev != -1) {
7         int diff = root->val - *prev;
8         if (diff < *ans)
9             *ans = diff;
10    }
11
12    *prev = root->val;
13    inorder(root->right, prev, ans);
14 }
15
16 int minDiffInBST(struct TreeNode* root) {
17     int prev = -1;
18     int ans = 1000000;
19
20     inorder(root, &prev, &ans);
21
22     return ans;
23 }
24
25 
```

Saved

Ln 20, Col 23

Testcase

Test Result

Lab Program:8a

Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm

Code:

```
#include <stdio.h>

#define MAX 10
#define INF 99999

int main() {
    int n;
    int graph[MAX][MAX];
    int selected[MAX] = {0};

    int edges = 0;
    int minCost = 0;

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter adjacency matrix (0 for no edge):\n");
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &graph[i][j]);

    selected[0] = 1;

    printf("\nEdges in MST:\n");

    while (edges < n - 1) {

        int min = INF;
        int x = -1, y = -1;

        for (int i = 0; i < n; i++) {

            if (selected[i]) {

                for (int j = 0; j < n; j++) {

                    if (!selected[j] &&
                        graph[i][j] &&
                        graph[i][j] < min) {
```

```

        min = graph[i][j];
        x = i;
        y = j;
    }
}
}

printf("%d - %d : %d\n", x, y, graph[x][y]);

minCost += graph[x][y];
selected[y] = 1;
edges++;
}

printf("\nMinimum Cost = %d\n", minCost);

return 0;
}

```

Output:

Output	Clear
<pre> Enter number of vertices: 5 Enter adjacency matrix (0 for no edge): 0 2 0 6 0 2 0 3 8 5 0 3 0 0 7 6 8 0 0 9 0 5 7 9 0 Edges in MST: 0 - 1 : 2 1 - 2 : 3 1 - 4 : 5 0 - 3 : 6 Minimum Cost = 16 === Code Execution Successful === </pre>	

Lab Program:8b

Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

Code:

```
#include <stdio.h>

#define MAX 10

struct Edge {
    int u, v, w;
};

int parent[MAX];

// Find parent
int find(int i) {
    while (parent[i] != i)
        i = parent[i];
    return i;
}

void unite(int a, int b) {
    parent[a] = b;
}

void sort(struct Edge e[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (e[j].w > e[j + 1].w) {
                struct Edge temp = e[j];
                e[j] = e[j + 1];
                e[j + 1] = temp;
            }
        }
    }
}

int main() {
    int n;
    int graph[MAX][MAX];
    struct Edge edges[MAX * MAX];

    int edgeCount = 0;
```

```

int minCost = 0;

printf("Enter number of vertices: ");
scanf("%d", &n);

printf("Enter adjacency matrix (0 for no edge):\n");

for (int i = 0; i < n; i++) {
    parent[i] = i;

    for (int j = 0; j < n; j++) {
        scanf("%d", &graph[i][j]);

        if (graph[i][j] != 0 && i < j) {
            edges[edgeCount].u = i;
            edges[edgeCount].v = j;
            edges[edgeCount].w = graph[i][j];
            edgeCount++;
        }
    }
}

sort(edges, edgeCount);

printf("\nEdges in MST:\n");

int count = 0;

for (int i = 0; i < edgeCount && count < n - 1; i++) {

    int u = find(edges[i].u);
    int v = find(edges[i].v);

    if (u != v) {
        printf("%d - %d : %d\n",
            edges[i].u,
            edges[i].v,
            edges[i].w);

        minCost += edges[i].w;
        unite(u, v);
        count++;
    }
}

printf("\nMinimum Cost = %d\n", minCost);

```

```
    return 0;  
}
```

Output:

Output

[Clear](#)

```
Enter number of vertices: 4  
Enter adjacency matrix (0 for no edge):  
0 2 0 6  
2 0 3 8  
0 3 0 5  
6 8 5 0
```

Edges in MST:

```
0 - 1 : 2  
1 - 2 : 3  
2 - 3 : 5
```

Minimum Cost = 10

=== Code Execution Successful ===

Lab Program:9

Implement Fractional Knapsack using Greedy technique

Code:

```
#include <stdio.h>

struct Item {
    int weight, value;
    float ratio;
};

void sort(struct Item arr[], int n) {
    for (int i = 0; i < n-1; i++)
        for (int j = 0; j < n-i-1; j++)
            if (arr[j].ratio < arr[j+1].ratio) {
                struct Item temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
}

int main() {
    int n, capacity;
    struct Item arr[100];

    printf("Enter number of items: ");
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        printf("\nItem %d\n", i+1);
        printf("Value: ");
        scanf("%d", &arr[i].value);
        printf("Weight: ");
        scanf("%d", &arr[i].weight);
        arr[i].ratio = (float)arr[i].value / arr[i].weight;
    }

    printf("\nEnter knapsack capacity: ");
    scanf("%d", &capacity);

    sort(arr, n);

    float totalValue = 0.0;

    printf("\nItems taken:\n");
```

```

for (int i = 0; i < n; i++) {
    if (capacity == 0) break;

    if (arr[i].weight <= capacity) {

        capacity -= arr[i].weight;
        totalValue += arr[i].value;
        printf("Item %d -> FULL (Value = %d)\n", i+1, arr[i].value);
    } else {

        float fraction = (float)capacity / arr[i].weight;
        totalValue += arr[i].value * fraction;
        printf("Item %d -> %.2f fraction\n", i+1, fraction);
        capacity = 0;
    }
}

printf("\nMaximum Value = %.2f\n", totalValue);

return 0;
}

```

Output:

Output

Clear

```

Enter number of items: 5

Item 1
Value: 50
Weight: 50

Item 2
Value: 30
Weight: 10

Item 3
Value: 60
Weight: 70

Item 4
Value: 10
Weight: 5

Item 5
Value: 90
Weight: 25

Enter knapsack capacity: 120

Items taken:
Item 1 -> FULL (Value = 90)
Item 2 -> FULL (Value = 30)
Item 3 -> FULL (Value = 10)
Item 4 -> FULL (Value = 50)
Item 5 -> 0.43 fraction

Maximum Value = 205.71

=== Code Execution Successful ===

```

LeetCode Problem:11

You are given an integer array height of length n. There are n vertical lines drawn such that the two endpoints of the i^{th} line are (i, 0) and (i, height[i]).

Find two lines that together with the x-axis form a container, such that the container contains the most water.

Return *the maximum amount of water a container can store.*

Notice that you may not slant the container.

Code:

```
int maxArea(int* height, int heightSize) {
    int left = 0;
    int right = heightSize - 1;
    int maxWater = 0;

    while (left < right) {
        int width = right - left;

        int h = (height[left] < height[right]) ? height[left] : height[right];

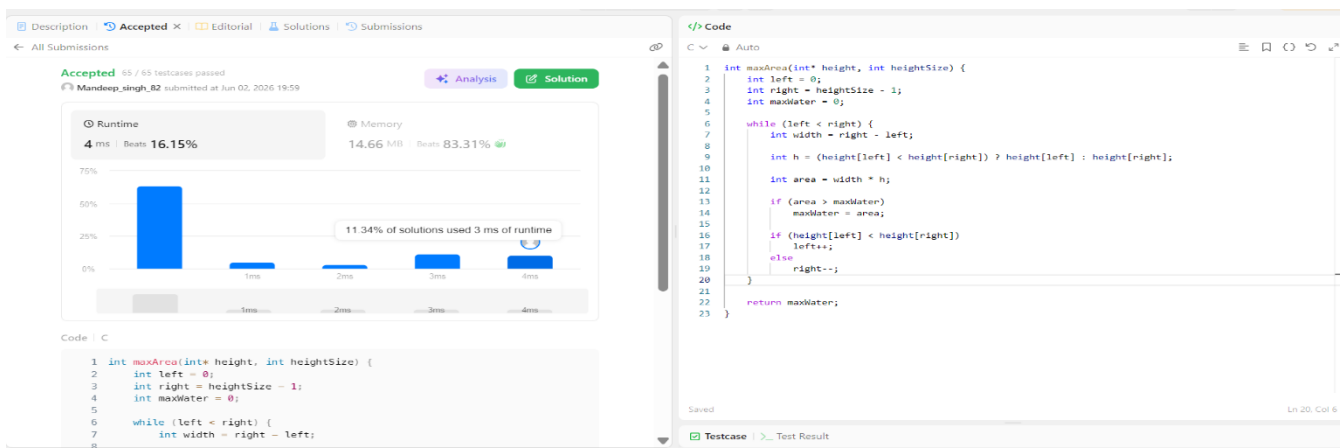
        int area = width * h;

        if (area > maxWater)
            maxWater = area;

        if (height[left] < height[right])
            left++;
        else
            right--;
    }

    return maxWater;
}
```

Output:



Lab Program:10

From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm

Code:

```
#include <stdio.h>
#define MAX 100
#define INF 99999

int minDist(int dist[], int visited[], int n) {
    int min = INF, index = -1;

    for (int i = 0; i < n; i++) {
        if (!visited[i] && dist[i] < min) {
            min = dist[i];
            index = i;
        }
    }
    return index;
}

void dijkstra(int graph[MAX][MAX], int n, int src) {
    int dist[MAX], visited[MAX];

    for (int i = 0; i < n; i++) {
        dist[i] = INF;
        visited[i] = 0;
    }
    dist[src] = 0;

    for (int i = 0; i < n-1; i++) {
        int u = minDist(dist, visited, n);
        visited[u] = 1;

        for (int v = 0; v < n; v++) {
            if (!visited[v] && graph[u][v] &&
                dist[u] + graph[u][v] < dist[v]) {

                dist[v] = dist[u] + graph[u][v];
            }
        }
    }

    printf("\nVertex\tDistance from Source\n");
```

```

        for (int i = 0; i < n; i++)
            printf("%d\t%d\n", i, dist[i]);
    }

    int main() {
        int n, src;
        int graph[MAX][MAX];

        printf("Enter number of vertices: ");
        scanf("%d", &n);

        printf("Enter adjacency matrix (0 for no edge):\n");
        for (int i = 0; i < n; i++)
            for (int j = 0; j < n; j++)
                scanf("%d", &graph[i][j]);

        printf("Enter source vertex: ");
        scanf("%d", &src);

        dijkstra(graph, n, src);

        return 0;
    }

```

Output:

Output

Clear

```

Enter number of vertices: 5
Enter adjacency matrix (0 for no edge):
0 5 6 0 2
5 0 0 4 6
6 0 0 3 5
0 4 3 0 4
2 6 5 4 0
Enter source vertex: 0

```

```

Vertex  Distance from Source
0      0
1      5
2      6
3      6
4      2

```

=== Code Execution Successful ===

Lab Program:11

Implement “N-Queens Problem” using Backtracking

Code:

```
#include <stdio.h>
#include <stdlib.h>

int x[20], n;

int place(int k, int i) {
    for (int j = 1; j < k; j++)
        if (x[j] == i || abs(x[j] - i) == abs(j - k))
            return 0;
    return 1;
}

void queen(int k) {
    for (int i = 1; i <= n; i++) {
        if (place(k, i)) {
            x[k] = i;
            if (k == n) {
                for (int j = 1; j <= n; j++)
                    printf("%d ", x[j]);
                printf("\n");
            } else
                queen(k + 1);
        }
    }
}

int main() {
    printf("Enter n: ");
    scanf("%d", &n);
    queen(1);
    return 0;
}
```

Output:

Output

Clear

Enter n: 5

1 3 5 2 4

1 4 2 5 3

2 4 1 3 5

2 5 3 1 4

3 1 4 2 5

3 5 2 4 1

4 1 3 5 2

4 2 5 3 1

5 2 4 1 3

5 3 1 4 2

=== Code Execution Successful ===

LeetCode Problem:704

Given an array of integers nums which is sorted in ascending order, and an integer target, write a function to search target in nums. If target exists, then return its index. Otherwise, return -1.

You must write an algorithm with $O(\log n)$ runtime complexity.

Code:

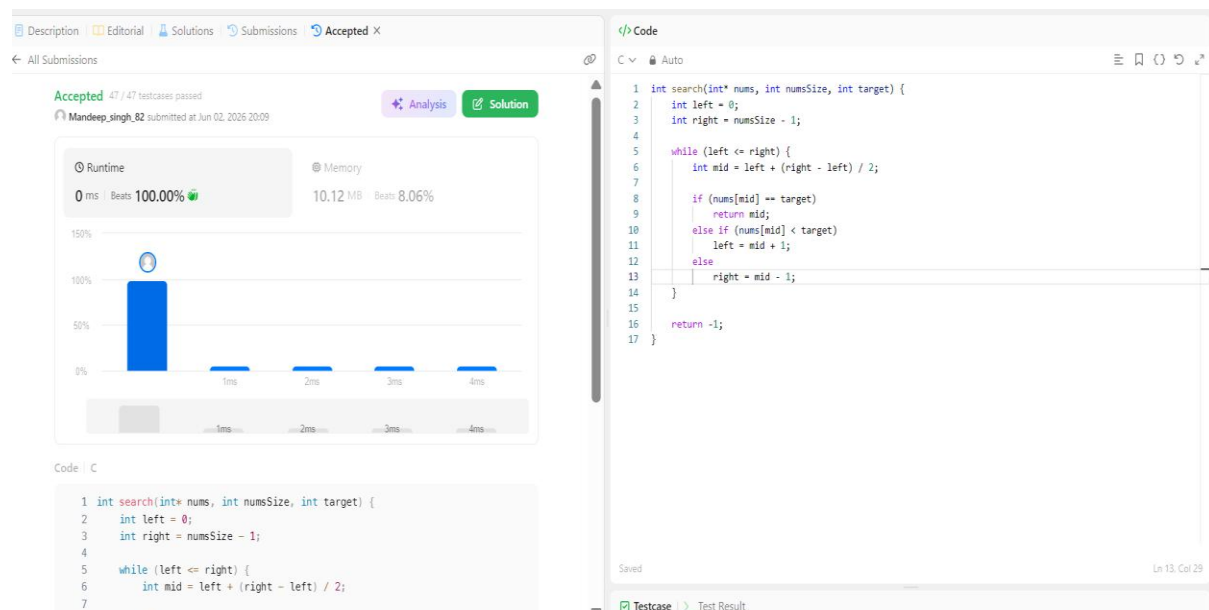
```
int search(int* nums, int numsSize, int target) {
    int left = 0;
    int right = numsSize - 1;

    while (left <= right) {
        int mid = left + (right - left) / 2;

        if (nums[mid] == target)
            return mid;
        else if (nums[mid] < target)
            left = mid + 1;
        else
            right = mid - 1;
    }

    return -1;
}
```

ScreenShot from LeetCode:



LeetCode Problem:268

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

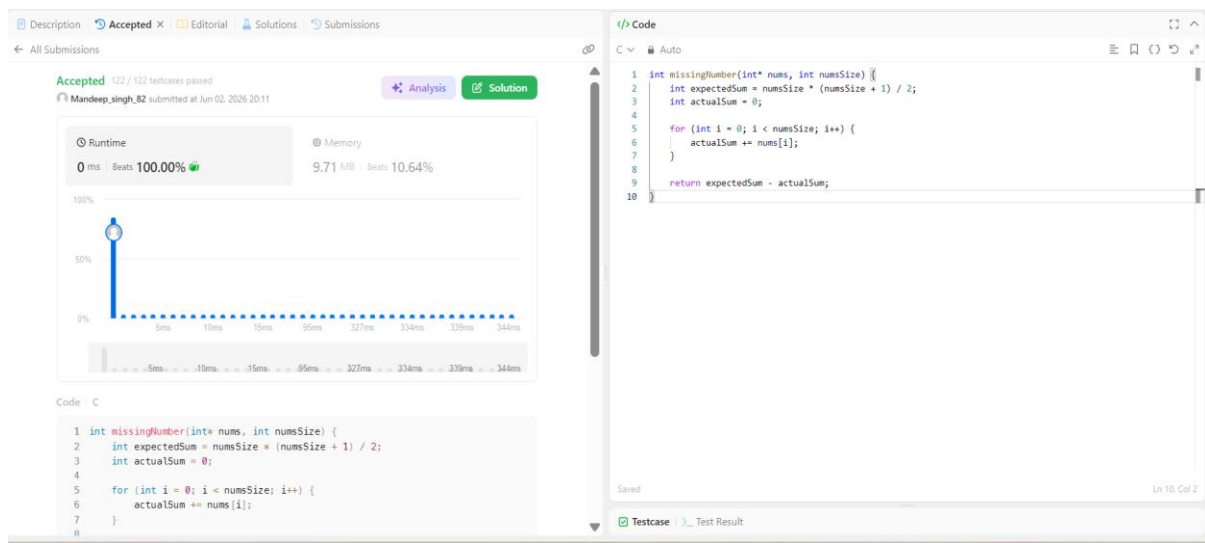
Code:

```
int missingNumber(int* nums, int numsSize) {
    int expectedSum = numsSize * (numsSize + 1) / 2;
    int actualSum = 0;

    for (int i = 0; i < numsSize; i++) {
        actualSum += nums[i];
    }

    return expectedSum - actualSum;
}
```

ScreenShot from LeetCode:



LeetCode Problem:1636

Given an array of integers `nums`, sort the array in **increasing** order based on the frequency of the values. If multiple values have the same frequency, sort them in **decreasing** order. Return the *sorted array*.

Code:

```
#include <stdlib.h>

typedef struct {
    int value;
    int freq;
} Pair;

int compare(const void *a, const void *b) {
    Pair *p1 = (Pair *)a;
    Pair *p2 = (Pair *)b;

    if (p1->freq != p2->freq)
        return p1->freq - p2->freq;    // increasing frequency

    return p2->value - p1->value;    // decreasing value
}

int* frequencySort(int* nums, int numsSize, int* returnSize) {
    Pair arr[numsSize];

    for (int i = 0; i < numsSize; i++) {
        int count = 0;
        for (int j = 0; j < numsSize; j++) {
            if (nums[i] == nums[j])
                count++;
        }
        arr[i].value = nums[i];
        arr[i].freq = count;
    }
    qsort(arr, numsSize, sizeof(Pair), compare);

    int *result = (int *)malloc(numsSize * sizeof(int));

    for (int i = 0; i < numsSize; i++) {
        result[i] = arr[i].value;
    }

    *returnSize = numsSize;
    return result;
}
```

ScreenShot from LeetCode:

